

is shown in Figure 4 with the open dyslexic Chrome extension enabled. It can be observed that the characters have a stronger base, which makes reading easier for users with dyslexia.

For the mobility challenged

For people who have difficulty handling the keyboard and mouse, there are some specialised tools. Users who cannot handle the mouse can control their mouse pointer by head movements, which are captured via the Web camera. *Enable Viacam* is an open source, free software that doesn't need any additional hardware. *Enable Viacam* can be downloaded from <http://eviacam.sourceforge.net/index.php>. There are versions available for Windows and Linux distributions.

For people with severe motor disabilities, there is an extension called *Hawking Toolbar*, which is named after the famous physically challenged scientist, Stephen Hawking. The *Hawking Toolbar* enables those with severe motor disabilities to access the Internet. The extension presents the links in the Web page to users in a sequential manner, and users can choose their special devices.

An extensive list of open source accessibility tools can be found at: <http://www.oatsoft.org/Software/listing/Repository>

Web Content Accessibility Guidelines (WCAG 2.0)

Though the assistive tools render the contents in an accessible format, the basic content itself should be in a particular standard for enhanced accessibility. The World Wide Web Consortium (W3C) has recommended a standard for Web developers and designers so that their content can be accessible to everyone. The most recent standard is WCAG 2.0-<http://www.w3.org/TR/WCAG20/>. A website can adhere to the rules of WCAG 2.0 at different levels. Based on these levels, ratings like A, AA, etc., are ascribed to the site.

In India, the Ministry of Social Justice and Empowerment, Department of Disability Affairs, gives the 'Best Accessible Website' award to websites adhering to WCAG 2.0 rules.

Sonar: An accessible OS

To make the computer completely accessible, the role of the OS is critical. There is a special Linux distribution focused largely on accessibility called Sonar. Sonar Linux is targeted at disabled users and provides important accessibility tools under one roof. Sonar Linux is developed by Jonathan Nadeau, who is himself visually challenged. The major features of Sonar include the following:

- It includes a screen magnifier and screen reader (ORCA by default).
- For those whose motor skills are impaired, it provides a special keyboard for easier access.
- For dyslexic people, it has a special font that makes the reading of text simpler.

Sonar Linux 13.10 can be downloaded from <http://>

sonargnulinux.com/?page_id=47. It can be installed on either 32-bit or 64-bit machines.

Accessible computing has, in fact, really come a long way. There is still scope for a lot more to be done. Tools specific to our country require a big push in terms of regional language requirements, etc. Open source developers can play a great role in making an impact in the day-to-day life of people with special needs. Imagine the satisfaction a developer gets in providing a real-life solution to the needy. **END** 

References

- [1] <http://www.un.org/disabilities/default.asp?id=1607>
- [2] <http://sonargnulinux.com/>
- [3] <http://www.oatsoft.org/Software/listing/Repository>
- [4] <http://www.w3.org/TR/WCAG20/>

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